

MAT 322 HW05

Aditya Pahuja

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Problem 1. (a) Suppose $f, g: Q \rightarrow \mathbb{R}$ are Riemann integrable functions on a rectangle $Q \subseteq \mathbb{R}^n$. Prove that $f + g$ is also Riemann integrable and that $\int_Q f + g = \int_Q f + \int_Q g$.

(b) Show there exists a subset $S \subseteq \mathbb{R}^2$ with measure zero whose closure and boundary both do not have measure zero.

(c) Prove that non-empty open sets of $\mathbb{R}^n$ do not have measure zero.

(d) Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function. Prove that its graph

$$G_f = \{(x, y) \in [a, b] \times \mathbb{R} : y = f(x)\}$$

is a subset of $\mathbb{R}^2$ with measure zero.

Solution. (a) Given a partition $\mathcal{P}$ of Q , we see that

$$U(f, \mathcal{P}) + U(g, \mathcal{P}) = \sum_{R \in \mathcal{P}} \sup_{x \in R} f(x) + \sup_{x \in R} g(x) \geq \sum_{R \in \mathcal{P}} \sup_{x \in R} (f(x) + g(x)) = U(f + g, \mathcal{P})$$

where the sums are over all rectangles in the partition; similarly we have the reverse inequality

$$L(f, \mathcal{P}) + L(g, \mathcal{P}) \leq L(f + g, \mathcal{P})$$

for the lower Riemann sums. Thus we have that for any partition $\mathcal{P}$ of Q ,

$$L(f, \mathcal{P}) + L(g, \mathcal{P}) \leq L(f + g, \mathcal{P}) \leq U(f + g, \mathcal{P}) \leq U(f, \mathcal{P}) + U(g, \mathcal{P}).$$

If $U(f, \mathcal{P}) - L(f, \mathcal{P})$ and $U(g, \mathcal{P}) - L(g, \mathcal{P})$ are both less than $\varepsilon/2$ (which we can arrange with a suitable choice of $\mathcal{P}$ due to Riemann integrability), then $U(f + g, \mathcal{P}) - L(f + g, \mathcal{P}) < \varepsilon$, implying $f + g$ is Riemann integrable over Q . Furthermore, given $\varepsilon > 0$, if $\mathcal{P}$ is chosen so that

$$U(f, \mathcal{P}) - \frac{\varepsilon}{2} < \int_Q f < L(f, \mathcal{P}) + \frac{\varepsilon}{2}, \quad U(g, \mathcal{P}) - \frac{\varepsilon}{2} < \int_Q g < L(g, \mathcal{P}) + \frac{\varepsilon}{2},$$

then the upper and lower Riemann sums of $f + g$ on $\mathcal{P}$ are bounded between $\int_Q f + \int_Q g - \varepsilon$ and $\int_Q f + \int_Q g + \varepsilon$, so that

$$\int_Q f + \int_Q g - \varepsilon \leq \int_Q f + g \leq \int_Q f + \int_Q g + \varepsilon$$

for all $\varepsilon > 0$ implies that

$$\int_Q f + \int_Q g = \int_Q f + g$$

as desired.

(b) Take $S = \mathbb{Q}^2$. Since S is countable, it has measure zero, and its interior is empty because any open ball contains points with all coordinates irrational, so the closure and boundary of S are both all of $\mathbb{R}^2$. Clearly $\mathbb{R}^2$ does not have measure zero.

(c) We first show that rectangles Q with nonzero volume do not have measure zero. Let $Q_1, Q_2, \dots$ be a countable collection of rectangles whose union contains Q . Then, let $\varepsilon > 0$ be any real

number. Grow Q_n into an open rectangle $R_n \supseteq Q_n$ whose volume exceeds Q_n by at most $\varepsilon/2^n$, so that

$$\sum_{n=1}^{\infty} \text{vol}(R_n) \leq \sum_{n=1}^{\infty} \text{vol}(Q_n) + \frac{\varepsilon}{2^n} = \varepsilon + \sum_{n=1}^{\infty} \text{vol}(Q_n).$$

Now the R_n form an open cover of the compact set Q , so some finite subcollection of the R_n covers Q , say $R_{i_1}, \dots, R_{i_k}$. It is easy to show that

$$\text{vol}(Q) \leq \text{vol}\left(\bigcup_{j=1}^k R_{i_j}\right) \leq \sum_{j=1}^k \text{vol}(R_{i_k}),$$

so we can conclude that

$$\text{vol}(Q) \leq \varepsilon + \sum_{n=1}^{\infty} \text{vol}(Q_n)$$

for all $\varepsilon > 0$, which implies that the total volume of the Q_n is at least as large as that of Q , hence the total volume of the Q_n is bounded below by a positive real number, which means Q cannot have measure zero.

Now, let $U \subseteq \mathbb{R}^n$ be open. We can find an open ball $B \subseteq U$, which in turn contains a rectangle Q of nonzero volume. Any covering of U with countably many rectangles Q_n must also cover Q , at which point the work above shows that the total volume of the Q_n is bounded below by a positive real number, so U also cannot have measure zero.

(d) Since $[a, b]$ is compact, f must be uniformly continuous. Given $\varepsilon > 0$, pick δ so that $|x - y| < \delta$ for $x, y \in [a, b]$ implies $|f(x) - f(y)| < \frac{\varepsilon}{b-a}$. Choose positive integer n so that $\frac{b-a}{n} < \delta$. Then, on each interval $[a + \frac{k(b-a)}{n}, a + \frac{(k+1)(b-a)}{n}]$ for integer $0 \leq k < n$, the maximum and minimum of f on the interval differ by less than $\frac{\varepsilon}{b-a}$, so that $G_f \cap ([a + \frac{k(b-a)}{n}, a + \frac{(k+1)(b-a)}{n}] \times \mathbb{R})$ can be covered with a rectangle with dimensions $\frac{b-a}{n} \times \frac{\varepsilon}{b-a}$ and volume $\frac{\varepsilon}{n}$. This implies that G_f can be covered by n such rectangles for a total volume of ε , hence G_f can be covered by an at most countable collection of rectangles with volume at most ε for any $\varepsilon > 0$, which is what it means for G_f to have measure zero. $\square$

Problem 2. (a) Show that if $S \subseteq \mathbb{R}^n$ is a compact subset of measure zero, then, for any $\varepsilon > 0$, S can be covered with finitely many rectangles with total volume less than ε .

(b) Let $Q \subseteq \mathbb{R}^n$ be a rectangle and let $f: Q \rightarrow \mathbb{R}$ be a bounded function. Show that if f vanishes outside a closed set of measure zero, then $\int_Q f$ exists and equals zero.

Solution. (a) By definition of S having measure zero, there exists a countable collection $Q_1, Q_2, \dots$ of rectangles whose union contains S and whose total volume is less than $\varepsilon/2$. For each n , let R_n be an open rectangle containing Q_n whose volume is at most $Q_n + \varepsilon/2^{n+1}$, so that the total volume of the R_n exceeds that of Q_n by at most $\varepsilon/2$, hence is less than ε . Then the R_n form an open cover of S , so we can find a finite collection $R_{i_1}, \dots, R_{i_k}$ such that their union contains S and their total volume is at most $\sum_{n=1}^{\infty} \text{vol}(R_n) < \varepsilon$. This suffices; if we want closed rectangles then we just replace each R_{i_j} with its closure, as a rectangle has the same volume as its interior.

(b) Since Z is closed and bounded, it is compact, so for a given $\varepsilon > 0$, it can be covered with finitely many boxes (which we will restrict to be contained in Q) whose total volume is at most ε , by part (a). Let $\mathcal{P}$ be the partition induced by these boxes (so $\mathcal{P}$ is the coarsest partition such that each of these boxes are unions of some rectangles in $\mathcal{P}$). If $m \leq f \leq M$ are lower and upper bounds for f on Q , then

$$m\varepsilon \leq L(f, \mathcal{P}) \leq U(f, \mathcal{P}) \leq M\varepsilon$$

since the total volume of the rectangles in $\mathcal{P}$ that intersect Z is at most ε by construction of $\mathcal{P}$, and all the other rectangles contribute zero to the Riemann sums. In particular

$$m\varepsilon \leq \int_Q f \leq \overline{\int}_Q f \leq M\varepsilon \implies \int_Q f = \overline{\int}_Q f = \int_Q f = 0$$

by taking $\varepsilon \rightarrow 0$, so f is Riemann integrable over Q and has integral zero. $\square$

Problem 3. (a) Let $f, g: S \rightarrow \mathbb{R}$ be bounded functions on a bounded set $S \subseteq \mathbb{R}^n$. Let

$$M(x) = \max(f(x), g(x)), \quad m(x) = \min(f(x), g(x)).$$

Show that if f and g are continuous (resp. Riemann integrable over S) then so are M and m .

(b) Let $S = S_1 \cup S_2 \subseteq \mathbb{R}^n$ be a bounded set, and $f: S \rightarrow \mathbb{R}$ be a function that is Riemann integrable over S_1 and S_2 . Prove that f is Riemann integrable over S and $S_1 \cap S_2$ and

$$\int_S f = \int_{S_1} f + \int_{S_2} f - \int_{S_1 \cap S_2} f.$$

Solution. (a) For continuity, we can write

$$\max(f(x), g(x)) = \frac{f(x) + g(x) + |f(x) - g(x)|}{2},$$

since if WLOG $f(x) \geq g(x)$ then

$$\frac{f(x) + g(x) + |f(x) - g(x)|}{2} = \frac{f(x) + g(x) + f(x) - g(x)}{2} = f(x) = \max(f(x), g(x)).$$

Linear combinations and compositions preserve continuity (noting that $x \mapsto |x|$ is continuous), so this is continuous because f and g are. Then since $\min(f(x), g(x)) = f(x) + g(x) - \max(f(x), g(x))$, $m(x)$ is a linear combination of continuous functions, hence continuous.

For Riemann integrability, we start by observing that if f and g are continuous at x , then so are m and M . If f and g are Riemann integrable over S , f and g are only discontinuous on a set of measure zero within some rectangle $Q \supseteq S$ (where we extend f and g to be zero outside S). The union of the sets of discontinuities of f and g has measure zero because the union of two measure zero sets has measure zero, and the set of discontinuities of M and m is a subset of this union, so M and m are also discontinuous on a set of measure zero, hence Riemann integrable over S .

(b) Note that we can write the positive part of f as $f_+ = \frac{f+|f|}{2}$, so f_+ is Riemann integrable on S_1 and S_2 . Similarly, so is the negative part $f_- = \frac{|f|-f}{2}$. Now, $f_+ = \max(f_{+S_1}, f_{+S_2})$, so by (a) f_+ is Riemann integrable on S (since the restricted functions are also Riemann integrable on S by virtue of being zero outside S_1 and S_2 respectively). Analogously, $f_{+S_1 \cap S_2} = \min(f_{+S_1}, f_{+S_2})$ implies that f_+ is Riemann integrable on $S_1 \cap S_2$. We can also conclude that f_- is integrable on S and $S_1 \cap S_2$, hence the linear combination $f_+ - f_- = f$ is integrable on S and $S_1 \cap S_2$.

Furthermore, we can rewrite

$$\int_E f = \int_S f \cdot 1_E$$

for any $E \subseteq S$ on which f is Riemann integrable, so proving the identity of integrals amounts to showing that

$$f = f \cdot 1_{S_1} + f \cdot 1_{S_2} - f \cdot 1_{S_1 \cap S_2}$$

by the linearity of the Riemann integral, and this identity is trivial, so the integral identity holds. $\square$

Problem 4. (a) Suppose $S_1, S_2 \subseteq \mathbb{R}^n$ are rectifiable sets. Prove that $S_1 \cup S_2$ and $S_1 \cap S_2$ are rectifiable, and their volumes are related by

$$\text{vol}(S_1) + \text{vol}(S_2) = \text{vol}(S_1 \cup S_2) + \text{vol}(S_1 \cap S_2).$$

(b) Suppose $S \subseteq \mathbb{R}^n$ is a bounded set which is a union of countably many rectifiable sets $S_1, S_2, S_3, \dots$. Prove or disprove that S must be rectifiable.

Solution. (a) Note that a bounded set S is rectifiable if and only if there exists a rectangle $Q \supseteq S$ such that the function $1_S: Q \rightarrow \mathbb{R}$ defined by

$$1_S(x) := \begin{cases} 1 & x \in S \\ 0 & x \notin S \end{cases}$$

is Riemann integrable over S , as the discontinuities of 1_S are precisely the points on the boundary of S , and Riemann integrability is equivalent to this set of discontinuities having measure zero. Furthermore,

$$\text{vol}(S) = \int_S 1_S.$$

Problem 3(b) now applies: we know that $1_{S_1 \cup S_2}$ is Riemann integrable over S_1 and S_2 , so it must be Riemann integrable over $S_1 \cup S_2$ and $S_1 \cap S_2$ as well, with

$$\begin{aligned} \text{vol}(S_1 \cup S_2) &= \int_{S_1 \cup S_2} 1_{S_1 \cup S_2} \\ &= \int_{S_1} 1_{S_1 \cup S_2} + \int_{S_2} 1_{S_1 \cup S_2} - \int_{S_1 \cap S_2} 1_{S_1 \cup S_2} \\ &= \text{vol}(S_1) + \text{vol}(S_2) - \text{vol}(S_1 \cap S_2), \end{aligned}$$

which rearranges to the desired expression for the sum of the volumes of S_1 and S_2 .

(b) S need not be rectifiable. For example, consider an enumeration $(q_n)_{n \in \mathbb{N}}$ of $\mathbb{Q} \cap [0, 1]$ and define $S_n = \{q_1, q_2, \dots, q_n\}$. Each S_n is a finite set, so the boundary of S_n is S_n which clearly has measure zero (given $\varepsilon > 0$, we can just place a rectangle of volume ε/n on each point of S_n); that is, the S_n are all rectifiable. However, the union of the S_n is $S = \mathbb{Q} \cap [0, 1]$, which isn't rectifiable because the closure of S is $[0, 1]$ while the interior is empty, so that the boundary of S is $[0, 1]$, which clearly doesn't have measure zero. $\square$

Problem 5. In this question we only consider integrals in the “extended sense.”

(a) Prove that $\int_{\mathbb{R}^2} e^{-x^2-y^2}$ exists and equals π .

(b) Prove that the **Gaussian integral** $\int_{\mathbb{R}} e^{-x^2}$ exists and equals $\sqrt{\pi}$.

Solution. (a) Consider the increasing sequence $K_1 \subseteq K_2 \subseteq \dots \subseteq \mathbb{R}^2$ of compact rectifiable sets where K_n is the circle of radius n centered at $(0, 0)$. The union of the K_n is all of $\mathbb{R}^2$. Furthermore, by a change of coordinates, we can compute

$$\int_{K_n} e^{-x^2-y^2} = \int_{(0, 2\pi) \times (0, n)} e^{-r^2} \cdot r,$$

where r (which varies across $(0, n)$) is the determinant of the Jacobian of the diffeomorphism that changes Cartesian coordinates to polar coordinates (actually the RHS is the integral over K_n minus its boundary and the positive x -axis, but these sets contribute zero to the integral anyway). The inner integral exists independently of the first coordinate because it is continuous in r , so Fubini's theorem gives

$$\int_{(0, 2\pi) \times (0, n)} e^{-r^2} \cdot r = \int_{\theta \in (0, 2\pi)} \left(\int_{r \in (0, n)} r e^{-r^2} \right) = \pi \cdot (1 - e^{-n^2}).$$

This means

$$\int_{\mathbb{R}^2} e^{-x^2-y^2} = \lim_{n \rightarrow \infty} \int_{K_n} e^{-x^2-y^2} = \pi < \infty,$$

so the integral over $\mathbb{R}^2$ exists and equals π .

(b) This integral exists because, noting that e^{-x^2} is continuous on $\mathbb{R}$, it is Riemann integrable on $[-n, n]$ for all n . Furthermore, we can observe the inequality

$$e^{-m^2} \leq e^{-m} \leq 2^{-n}$$

for all nonnegative integers m . If $m(x)$ is the floor of $|x|$, this lets us bound $e^{-x^2} \leq 2^{-m(x)}$. We can calculate for all positive integers n

$$\int_{[-n, n]} 2^{-m(x)} = 2 \sum_{k=0}^{n-1} 2^{-k} = 2(2 - 2^{-n+1})$$

(integrability follows from being piecewise constant, so $2^{-m(x)}$ has finitely many discontinuities) so taking the limit of n shows that $2^{-m(x)}$ is integrable on $\mathbb{R}$ with integral equal to 4. Thus

$$\lim_{n \rightarrow \infty} \int_{[-n,n]} e^{-x^2} \leq \lim_{n \rightarrow \infty} \int_{[-n,n]} 2^{-m(x)} = 4 < \infty$$

implies that e^{-x^2} is integrable over $\mathbb{R}$. Now we can use Fubini's theorem to get

$$\int_{[-n,n] \times [-n,n]} e^{-x^2-y^2} = \int_{[-n,n]} e^{-x^2} \cdot \int_{[-n,n]} e^{-y^2} = \left(\int_{[-n,n]} e^{-x^2} \right)^2,$$

so that taking $n \rightarrow \infty$ gives

$$\pi = \int_{\mathbb{R}^2} e^{-x^2-y^2} = \left(\int_{\mathbb{R}} e^{-x^2} \right)^2,$$

which shows that the Gaussian integral is $\sqrt{\pi}$. □